



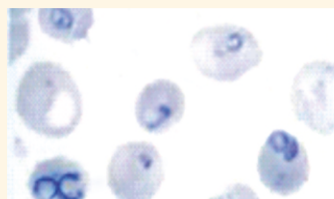
- **FTA-ABS**
- VDRL is done for diagnosis of neurosyphilis and response to treatment
- Peripheral smear

- Leading cause of transfusion-associated hepatitis (TAH): **Hepatitis B virus**
- Sexual: due to infected body fluids
- Perinatal: due to vertical transmission

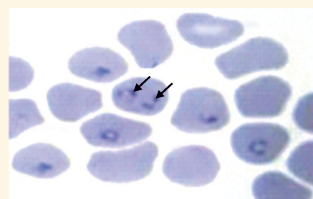
INCIDENCE

Fulminant hepatic failure incidence (Development of hepatic encephalopathy and/or coagulopathy)	0.1-1%
Progression to chronicity	1-10%
High carrier stage in which age group	90% of neonates
Cancer risk (HCC)	0.1-30%
Prognosis	Worse with age

Important Information



Maltese Cross App



P. Falciparum

- Peripheral smear: P. falciparum → **Multiple ring stage** + banana shaped gametocytes
 - Treatment: Artesunate
- **Maltese cross** (Babesia microti transmitted by Ixodes) appearance on peripheral smear: differential diagnosis of Falciparum malaria
 - Treatment: drug- Atovaqone and Azithromycin

WORKUP OF ETIOLOGY OF ACUTE HEPATITIS

00:37:28

HBSAG	IGM ANTI-HAV	IGM ANTI-HBC	ANTI-HCV	INTERPRETATION
+	-	+	-	Acute hepatitis B
-	+	-	-	Acute hepatitis A
-	-	-	+	Acute Hepatitis C
-	-	-	-	Non-A, Non-B, Non-C (hepatitis D, E)



• Presentation with acute hepatitis

- Right upper quadrant pain
- ↑ liver span on examination
- LFT deranged
- Hepatitis D occurs as co-infection or superinfection, it does not occur independently

PARAMETER	HEPATOCELLULAR JAUNDICE	OBSTRUCTIVE JAUNDICE	HAEMOLYTIC JAUNDICE
Total bilirubin	↑	↑	↑
Direct (conjugated)	↑	↑↑	N/↑
Indirect (unconjugated)	↑	Normal or ↑	↑↑
SGOT (AST)	↑↑	N/↑	N
SGPT (ALT)	↑↑↑	N/↑	N
ALP (SAP)	↑	>4 Times rise	N
Urine color due to bilirubinemia	Mustard yellow	Mustard yellow ++	Normal
Urine Urobilinogen	Low	Absent	↑
Stool color	Normal or pale	Pale/ clay	Normal

- Hemolysis: RBC breakdown → Formation of unconjugated bilirubin
- Bile canaliculi damage can be present in hepatocellular jaundice
- In obstructive jaundice:
 - Urobilinogen is produced from conjugated bilirubin in the intestine. → No conjugated bilirubin due to physical obstruction. → No conjugated bilirubin in the gut
 - Gut bacteria convert conjugated bilirubin into urobilinogen
- In **hemolytic jaundice**:
 - Unconjugated bilirubin ↑ → converted to conjugated bilirubin → more conjugated bilirubin in gut → bacterial flora acts on it → converted to urobilinogen → excreted in urine

CAUSES

CAUSES	JAUNDICE
Obstructive jaundice	• Carcinoma head of pancreas • Primary sclerosing cholangitis • Primary biliary cirrhosis
Hepatocellular jaundice	• Hepatitis B • Paracetamol toxicity



Hemolytic jaundice

- Malaria
- Hereditary spherocytosis

00:45:36

ACUTE HEPATITIS

- 90% spontaneous recovery
- Conservative management
- No antivirals needed unless there is risk for fulminant failure

CHRONIC HEPATITIS

- HBeAg positive
 - Threshold for treatment: PCR DNA $>2 \times 10^4$ IU/ml with SGPT doubling
- HBeAg negative
 - PCR DNA $>2 \times 10^3$ IU/ml with SGPT doubling (lower threshold)
- Treatment
 - Tenofovir with a watch on KFT
 - Entecavir

IMMUNE PHASE OF CHRONIC HBV

	IMMUNE TOLERANT	IMMUNE-ACTIVE	IMMUNE-CLEARANCE	REACTIVATION
ALT/SGPT	Low	High	Moderate	Low/moderate
HBV DNA	High	Moderate	Low	Low/moderate
HBeAg	+	+/-	-/+	-/+
Anti-HBe	-	-/+	+/-	+/-
Anti-HBV immune control	Weak	Strongest	Strongest	Weak
Need for treatment	No	Yes	No	Yes

HEPATITIS A

00:49:01

- Chances of fulminant hepatic failure is 0.1%-1%
- Fulminant hepatic failure can occur in all hepatitis virus from HAV to HEV (highest chance in pregnancy is with HEV)
- Carrier and chronic hepatitis are seen in all except HAV and HEV
- More symptomatic in adults
- Ig M Anti-HAV
- Inactivated vaccine available
 - Given 2 shots to children after 1 year of age, subcutaneous
- For acute infection, the class of antibody checked is IgM class of antibody to HAV
- No treatment required, self-limiting infection

HEPATITIS E

00:50:28

- M/c cause of acute viral hepatitis

- **Fulminant hepatic failure** in pregnancy

• **41-50% HDV**

- **Vaccines** Available in China

- Complete clinical and biochemical recovery is expected in all cases of HAV, HEV in 1-2 months of post-jaundice

TRIGGERS FOR HEPATIC ENCEPHALOPATHY

00:51:21

- GI bleed: contributes to **↑ ammonia** concentration
 - Liver failure → coagulopathy and bleeding
 - Maximum ammonia is produced in the body by intestinal bacteria and detoxified in the liver
 - Deranged Coagulogram (fulminant hepatic failure) → **Internal GI Bleeding** → acts as a good culture media → multiplication of gut flora → ammonia ↑ → trigger encephalopathy
- Diuretic excess
 - Loss of water → **Ammonia concentration ↑**
- Loss of potassium
- Dehydration
- Infection
- Constipation: Bacteria in intestine ↑
- Narcotics

EVALUATION OF LFT

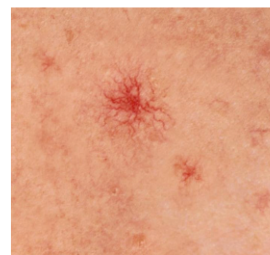
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- Hepatocellular Jaundice:
 - SGPT ↑
 - SGOT ↑
 - Serum Alkaline phosphatase ↑
 - Tests done are:
 - IgM Anti-HAV
 - HBsAg, IgM Anti Hbc
 - Anti-HCV
 - ANA, SMA: to detect autoimmune hepatitis
 - Ceruloplasmin, alcohol history, drug history
- Obstructive jaundice: tests done are
 - USG
 - MRCP
 - AMA (Anti-mitochondrial antibody): for diagnosis of **Primary Biliary Cholangitis**
 - Drug intake

FINDINGS IN ALCOHOLIC CIRRHOSIS

00:55:00

- **Spider angioma**: Dilated cutaneous arterioles looking like arterioles,
 - Hormone responsible is Estrogen
- **Palmar erythema**: feature of liver cell failure/compensated cirrhosis



Spider angiomas



Palmar erythema



Q. Comment on the diagnosis?

HBsAg	Negative
IgG anti-HBc	Present
HBeAg	Negative
Anti HBs	Negative

- a. Gap period
- b. Remote infection
- c. Recovery
- d. Co-infection

Ans. (b)



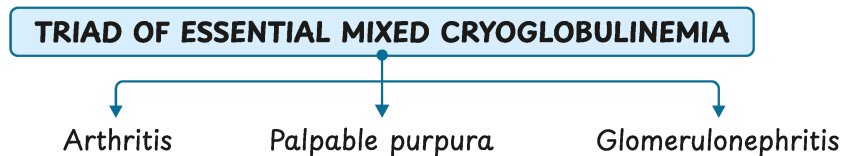
4. HEPATITIS C-D AND FULMINANT HEPATIC FAILURE



HEPATITIS C

EXTRAHEPATIC FEATURES

- MPGN: Membrano proliferative glomerulonephritis
- Essential mixed cryoglobulinemia (Antibodies which precipitate at 4°C at bottom of test tube and when temperature ↑ to 37°C, antibodies disappear)

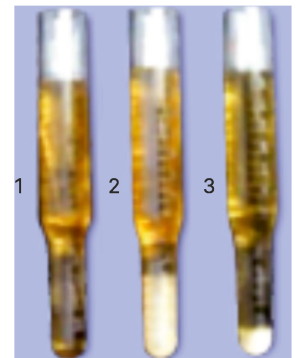


- Lichen Planus: purple polygonal plaques present on flexor areas
- Porphyria cutanea tarda - presence of photosensitivity
- Type 2DM, Metabolic syndrome
- Sicca syndrome and cutaneous vasculitis

FEATURES AND SPREAD OF VIRUS

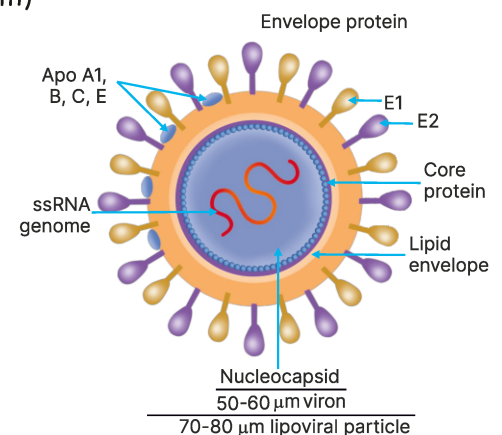
- **Largest viral particle** - Hepatitis C (55nm) (Dane particle size - 42 nm)
- Genus: Hepacivirus(RNA virus)
- Genome: 9.4kb, positive sense single stranded RNA virus
- Antigen: HCV core antigen
- Screening test: **EIA** (Enzyme Immunoassay anti HCV antibody) → may take couple of weeks to become positive
- **IOC: quantitative PCR HCV RNA**
- Chances of transmission due to needle stick injury: 3%
 - Chances of transmission due to needle stick injury in HBV: 30%
 - Chances of transmission due to needle stick injury in HIV: 0.3%
- Total number of genotypes - 6, **M/c** genotype in USA - 1
- 1st investigation for work up of accidental needle stick injury: q PCR HCV RNA
- Cross reactivity between NS3 NS5A & host cytochrome P 450 explains association with autoimmune hepatitis - Anti LKM1 antibody, histopathological findings- interface hepatitis and rosette formation
- Routes of spread
 - Percutaneous
 - May or may not be transmitted Perinatally
 - Sexual ±
 - Not transmitted by Feco-oral route
- Hepatitis C → 85% → Chronic hepatitis → 25% (stellate cells activated) → Cirrhosis/ End stage liver disease

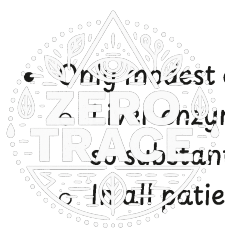
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Day 0 7 Centrifugation +4 °C

00:04:30



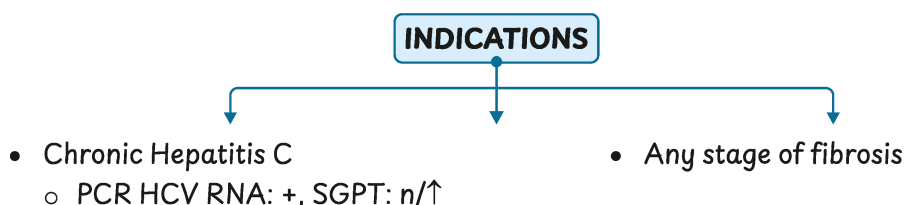


- Only modest elevation of aminotransferases seen (its like fire burning below surface, without smoke)
- Liver enzymes can be normal inspite of the virus causing cytopathic effect, rise of aminotransferases not so substantial as with other hepatotropic viruses
- In all patients irrespective of SGPT values, if virus present in blood- needs to be treated

MANAGEMENT

00:14:33

- Oral, highly effective, short duration directly acting antiviral agents (DAA)
- Objective:
 - Sustained virological remission (SVR₁₂) → Virus undetectable >12 weeks of T/t



T/t

- For genotypes 1a, 1b, 2, 3, 4, 5, 6
- Drugs: DOC - Sofosbuvir (NS5B-) +Velpatasvir (NS5A-)
 - MOA: NS5B- (inhibits RNA dependent RNA polymerase)
- Protease inhibitor: Glecaprevir (NS 3/4A protease inhibitor) + Pibrentasvir

ONE LINERS

- M/c cause of acute viral Hepatitis: HEV
- M/c cause of chronic Viral Hepatitis: HCV
- M/c cause of Chronic liver disease: Non-alcoholic fatty liver disease> Alcoholic hepatitis> hepatitis C
- M/c cause of Chronic liver diseases Needing liver Transplantation: NAFLD
- Largest gene of Hepatitis B virus: P gene
- Leading Cause of Transfusion-associated Hepatitis: HBV
- Leading cause of fulminant hepatic failure (F.H.F): Toxins >Hepatitis B
- Viral hepatitis with highest chances of F.H.F in pregnancy - HEV

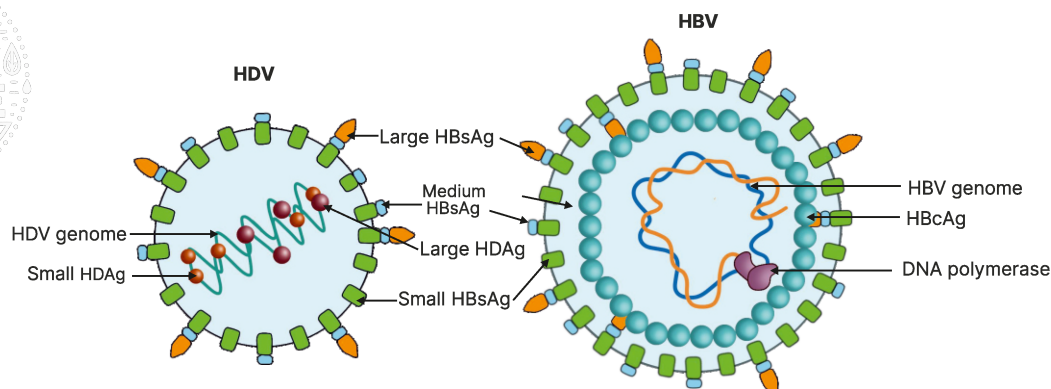
HEPATITIS D

00:20:34

- Enveloped hybrid particle HBsAg coat and HDV core
- Incomplete ssRNA virus that borrows surface antigen from Hepatitis B virus to infect new hepatocytes
- Eradication: 100% coverage of entire population with hepatitis B vaccine

PATTERNS OF INFECTION

COINFECTION (IV drug user)	• IgM anti HBc • HBsAg	Anti HDV (IgM)	FHF risk ↑ - 5-20%
SUPERINFECTION	• IgG Anti HBc • HBsAg • Chronic hepatitis B	Anti HDV	Chronic hepatitis risk ↑



HEPATITIS D STAGES AND SEROLOGICAL MARKERS - 3 INFECTIVE PATTERN

DIAGNOSTIC MARKERS	ACUTE HBV/HDV COINFECTION	ACUTE HDV SUPERINFECTION	CHRONIC HDV INFECTION
HD Ag	Transiently positive early	Transiently positive early	Negative
IgM Anti-HDV	Transiently positive late	Positive	Positive
IgG Anti-HDV	Positive late	Positive late	Positive
HDV RNA	Positive	Positive	Positive
IgM anti-HBC	Positive	Negative	Negative

ROUTE OF SPREAD

- Percutaneous, Perinatal, Sexual
- Fulminant hepatic failure: 5-20% risk with co-infection
- Progression to chronicity+
- T/t - Pegylated interferon **S/C** injections

HEPATITIS A AND E

00:28:40

	HAV	HEV
Age	Child, young adult	Young adult (epidemics during rainy season)
Transmission		
Feco-Oral	+	+
Percutaneous	Unusual	-
Sexual	+/-	-
Fulminant hepatic failure	0.1%	1%
Carrier	-	-
Chronic Hepatitis	-	-



vaccine

- Available, Immunoglobulin (passive immunity)
- >1 year age - 2 shots S/C (active immunity)

Available

- Fulminant hepatic failure seen in all viral hepatitis
- Carrier stage of chronic hepatitis not seen in HAV and HEV

NON-ALCOHOLIC FATTY LIVER DISEASE

00:32:01

- Leading indication for **liver transplantation**
- M/c cause of Chronic liver disease worldwide
- NAFLD - **Premalignant** condition
- NAFLD → NASH → Cirrhosis → HCC risk ↑
↓
Intrahepatic cholangiocarcinoma risk ↑

- Definition: >5% hepatocytes have accumulation of triglycerides
- NASH - Reversible by diet and lifestyle modification

OTHER CAUSES OF HEPATIC STEATOSIS

(Steatosis - Triglyceride accumulation in hepatocytes causing inflammation)

- Alcohol, Hepatitis C
- **Reye syndrome**: Aspirin + Varicella zoster (chickenpox)
 - Anicteric hepatitis → SGPT ↑
 - No jaundice but raised liver enzymes
 - Hepatic encephalopathy can occur
- Pregnancy, HELLP
- Metabolic syndrome, Wilson diseases
- Drugs
 - Tetracycline, Doxycycline, Amiodarone, Steroids

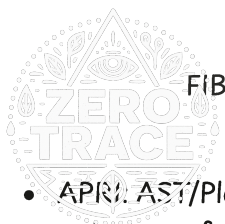
CLINICAL FEATURES

00:36:48

- Asymptomatic: Vague RUQ pain
- Hepatomegaly
- Centripetal Obesity -
 - Spider angioma: ↑ estrogen
 - Palmar erythema
 - Splenomegaly
- Association: OSA, PCOD

WORK UP

- Liver biopsy: Gold standard (invasive)
- USG (1st investigation): **Transient elastography** (imaging modality of choice- Fibroscan)
- FIB-4 score: Non invasive

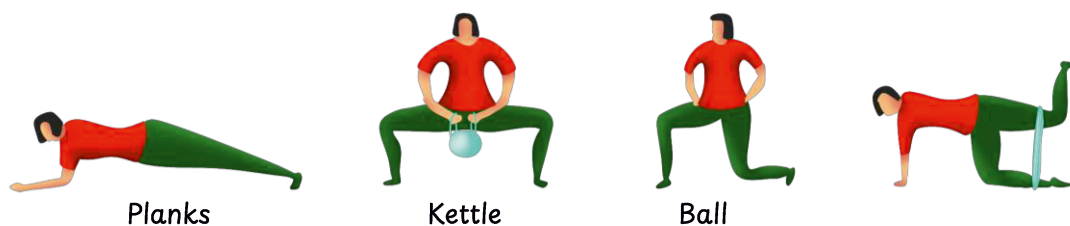


$$\text{FIB-4} = \frac{\text{Age (years)} \times \text{AST (U/L)}}{\text{Platelet Count } \left(\frac{10^9}{\text{L}}\right) \times \sqrt{\text{ALT (U/L)}}}$$

- **APRI: AST/Platelet**
 - In case of advanced liver diseases, platelet count of patient progressively ↓
- **BARD score: BMI, SGOT, SGPT, Diabetes Mellitus**
- **NAFLD Fibrosis Score**

T/t

- **Weight loss: 7-10%: Most effective**
 - Resistance training/aerobic exercises
- **Mediterranean diet**
- **Statins for deranged lipid profile**
- **New drug: Resmetirom** -Thyroid hormone receptor beta agonist approved for NASH with F2-F4 liver cirrhosis



MISCELLANEOUS POINTS

- **Zieve syndrome: Alcoholic hepatitis + Hemolytic anaemia (with peripheral smear findings (burr cell))**
- **Vitamin E long-term use causes worsening of CAD**
- **Initial screening test for suspected liver fat - USG abdomen**
- **Transient elastography - Advanced fibrosis: >7.3kPa, Cirrhosis: >15kPa**
- **Test of choice quantitative assessment of liver fat: MRI proton density fat fraction**

ACUTE LIVER INJURY

00:46:04

- **Hepatic encephalopathy -**
 - Ammonia ↑ → asterix development, aphasia, apraxia
 - Encephalopathy and/or
- **Coagulopathy - PT/INR ↑ (1.5 times) → GI bleeding**
 - Within 8 weeks of acute liver insult
 - Amanita poisoning, Pyrazinamide
 - Halothane, Valproate, PCM
- **If damage occurs between 8-26 weeks: Subfulminant hepatic failure**

CAUSES

- **Acetaminophen/PCM Toxicity**
- **ATT: Pyrazinamide (Most hepatotoxic)**
- **Antiepileptic drugs- Valproate**
- **Viral Hepatitis: HBV, HCV, HEV**
- **Mushroom poisoning: Amanita phalloides**

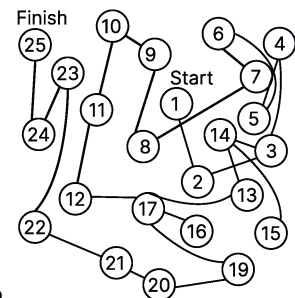
- Heat stroke
- GCS converting to Generalised Convulsive Status Epilepticus
- Luca Chiari syndrome, Wilson's disease, Herbal drugs

CLINICAL FEATURES

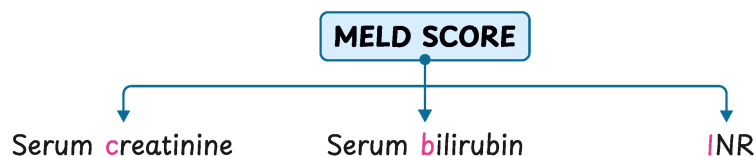
- Altered sleep pattern (**earliest** manifestation), inattention, euphoria, slurring voice
- Asterixis, Time disorientation
- Apraxia - Number connection test
- **Anomic aphasia** (present in metabolic encephalopathy) - Difficulty in naming an object
- Jaundice: May be minimal but gross elevations of Aminotransferase
- ↑ ICP, GI bleeding
- ↓ Urine output - Hepatorenal syndrome



Stage of HE	Time
0	Up to 30 s
0-I	31-50 s
I-II	51-80 s
II-III	81-120s
III	Forced termination



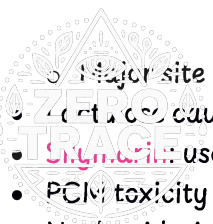
WORK UP



- LFT: ↑ bilirubin, ↑ SGPT, ↑ SGOT
- Blood ammonia ↑ (not reliable), West haven criteria used to diagnose and grade severity of hepatic encephalopathy
 - Includes clinical findings and EEG findings -
 - Asterixis with hyporeflexia (stage II of hepatic encephalopathy)
 - Asterixis with hyperreflexia (stage III of hepatic encephalopathy)
 - Disappearance of asterixis (severe diseases process of hepatic encephalopathy) and development of decerebrate posturing in an individual due to ↑ ICP
- EEG: **Triphasic waves**, Delta waves
- INR ↑
 - PT ↑, normal BT, normal aPTT
- Random blood sugar ↓

T/t

- Vasopressor: Norepinephrine
- I/V fluids: NS and 10% dextrose for sugar dips
- Neomycin/Rifaximin : **Bacteriostatic antibiotic** via NG tube to inhibit growth of GI flora
 - Major site for ammonia detoxification - liver



- Major site of production of ammonia in body - intestinal bacteria
- Lactulose causes loose stools: ↓ Flora in gut
- **Silymarin**: useful in mushroom poisoning → inhibits toxin uptake by hepatocytes
- PCM toxicity antidote- N/Acetylcysteine
- Nucleoside Analogs for HBV - Tenofovir used
- Raised ICP - Mannitol
- Refer to liver T/t centre - MELD score >30 (in case of emergency)
 - Liver transplantation done if MELD score >17
- **FFP** - Richest source of clotting factors
- LOLA Infusion - L-ornithine and L-aspartate

GRADES OF HEPATIC ENCEPHALOPATHY

00:59:22

GRADE	LEVEL OF CONSCIOUSNESS	PERSONALITY AND INTELLECT	NEUROLOGIC SIGNS	ELECTROENCEPH ALOGRAM (EEG)
Sub-Clinical	Normal	Normal	Abnormal only on psychometric testing	None
1	Day/night sleep reversal, restlessness	Forgetfulness mild confusion, agitation, irritability	Tremor, apraxia, incoordination, impaired handwriting	Triphasic waves (5 Hz)
2	Lethargy, slowed response	Disorientation to time, loss of inhibition, inappropriate behaviour	Asterixis, dysarthria, ataxia, hypoactive reflexes	Triphasic waves (5 Hz)
3	Somnolence, confusion	Disorientation to place, aggressive behaviour	Asterixis, muscular rigidity, Babinski signs , hyperactive reflexes	Triphasic waves (5 Hz)
4	Coma	None	Decerebration	Delta/Slow wave activity



Q. A 25-year-old man presents with fever, jaundice and right upper quadrant pain. He has no past history of liver diseases. Serologic test are as follows:

- HBsAg: Positive
- Anti HBc IgM: Positive
- Anti-HDV: Positive
- HDV RNA: Positive

Which of the following is the most likely diagnosis?

- a. Chronic hepatitis B infection
- b. Hepatitis B and D co-infection
- c. Hepatitis D superinfection
- d. Past resolved hepatitis B with recent HDV infection

Ans. b



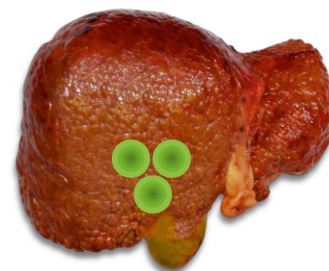
HEPATORENAL SYNDROME AND HEPATOPULMONARY SYNDROME

- Incidence: 10% cases of advanced cirrhosis (decompensated cirrhosis-portal hypertension manifestations)
- Type of Acute kidney injury seen: Pre-renal
- Histopathology findings: Kidney biopsy → Glomerulus normal
 - No parenchymal involvement of kidney, under perfusion occurs

PATHOPHYSIOLOGY

00:01:38

- Renal vasoconstriction
 - PGE_2 , PGI_2 → Causes **vasodilatation** of afferent arterioles of kidney → ↑ Renal blood flow → ↑ Glomerular filtration rate
 - In Hepatorenal syndrome: Urinary loss of Prostaglandins → **Renal vasoconstriction**
- Cirrhotic liver produces **Endothelin-1**
 - It activates **Endothelin Nitric oxide synthase** (eNOS) in splanchnic blood vessels
 - More production of NO will cause splanchnic vasodilatation and shunt blood away from kidney



REVISED CRITERIA FOR HRS

- **Refractory Ascites**: In spite of maximum amount of diuretics allowed for the individual, there is still presence of ascites
- **↑ S. Creatinine**
 - >50% increase over baseline within 7 days
 - >0.3 mg/dl over baseline within 48 hours
- Stop Diuretics for 2 days
 - Too much of diuretics → Loss water from body → Renal shutdown
- Volume expansion with **Albumin** → ↑ oncotic pressure → Fluid from peritoneal cavity goes back to blood vessels so that kidney perfusion can be maintained
- No shock
- No nephrotoxic drug: ACE inhibitors, NSAIDs
- No parenchymal disease (no proteinuria >500 mg/dl, no haematuria >50 RBC/HPF and normal imaging of kidney)

CASE SCENARIO

A 40-year-old alcoholic man with CLD and ascites developed a variceal bleed at home. In ER, BP 80/60 mmHg with HR 120/min and urine output is 100 ml in last 6 hours. Urine sodium is increased with urine osmolality of 200 mosm.

Diagnosis- **Acute tubular necrosis**

- Not HRS as there should be no circulatory failure
- Urinary sodium increases as lining of tubule is damaged



- Not every case of alcoholic cirrhosis with ascites will be HRS, if there is bleeding and hypotension → Acute tubular necrosis

TYPES

00:09:08

- HRS-AKI (Acute Kidney Injury)
- HRS-NAKI (Non-Acute Kidney Injury): Gradual worsening
- HRS-CKD
 - Already the kidney of the patient is compromised
 - $eGFR < 60$ ml/min

TREATMENT

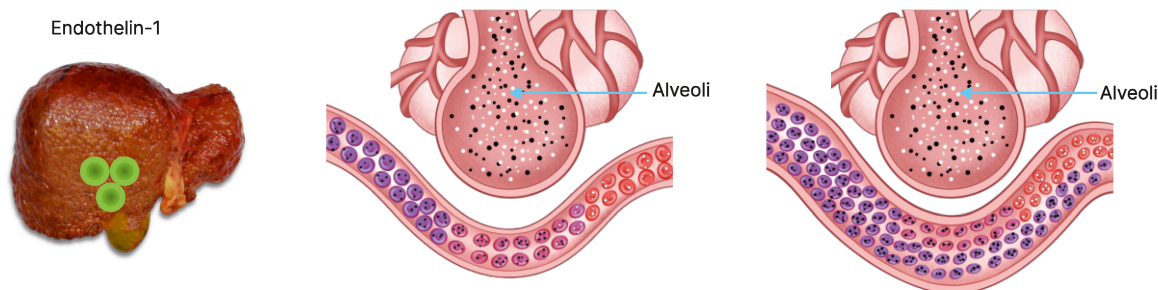
- Terlipressin and Albumin
 - Terlipressin is vasoconstrictor of splanchnic vessels by acting on $V1$ receptors and reduces splanchnic vasodilatation

OR

- Midodrine improves Systemic Vascular Resistance (SVR) + Octreotide (reduces portal pressure) and Albumin

HEPATOPULMONARY SYNDROME

00:11:19



- Cirrhotic liver → Endothelin 1 → $\uparrow$ eNOS → $\uparrow$ NO in pulmonary vasculature → Dilatation of pulmonary vessels
- Oxygen present in alveoli can easily diffuse in RBCs as these are close to blood vessels
- In **dilated blood vessel** Oxygen will be able to diffuse into some of the RBCs in the blood vessels but not in all RBCs d/t to dilation of the blood vessels
- Important feature of Hepatopulmonary syndrome is **Intrapulmonary shunting**

TRIAD OF HPS

- Intrapulmonary shunting
- Platypnea or Orthodeoxia :
 - Platypnea: Dyspnea worsening on sitting position
 - Orthodeoxia: $\downarrow$ oxygen saturation on change of position
 - Drop in oxygen saturation by **5%** on change of position from supine to sitting: Dilated blood vessels on sitting up due to gravity are pressing each other and adjacent lung → Breathing difficulty due to ventilation-perfusion imbalance
 - $\downarrow$ PaO_2 by **>4 mmHg**
 - $\uparrow$ Alveolar arteriolar gradient
 - $PaO_2 < 80$ mmHg
- Cirrhosis

Important Information

CAUSES OF PLATYPNEA

Atrial myxoma

- Emphysema (more air trapping on sitting) → bulla formation
- ARDS (more atelectasis on sitting)
- Hepatopulmonary syndrome

WORK UP

- IOC: Bubble contrast echocardiography
 - In normal individual: Agitated saline pushed through IV line → Right atrium, right ventricle, pulmonary circulation but not able to enter small pulmonary vessels and left side of heart
 - In HPS: Dilated blood vessels → Agitated saline can navigate across dilated blood vessels → Left atria of the patient



Bubble score = 0

Bubble score = 1

Bubble score = 2

Bubble score = 3

Bubble score = 4

MANAGEMENT

- Supplemental Oxygen
- Orthoptic liver transplant

MCQ

00:19:07

Q. A 40-year-old male with alcoholic cirrhosis presented to ER with complaints of breathing difficulty on sitting. No fluid thrill or shifting dullness is noted. Clinical examination revealed severe pallor and a fall in SpO_2 to 78-80% while sitting upright from a supine position. Which of the following is correct about this condition?

- a) Triad of intrapulmonary shunting + Platypnea + cirrhosis
 - b) Fall in A-a gradient
 - c) Investigation of choice is bubble contrast echocardiography
 - d) Terlipressin and albumin infusion
- a. a, b
b. a, c
c. b, c
d. a, d

Ans: (b)



13. AUTOIMMUNE HEPATITIS

- Formerly called: **Lupoid Hepatitis** or **Plasma cell hepatitis**
- In untreated autoimmune hepatitis, 6-month mortality rate: **40%**

PATHOPHYSIOLOGY

00:00:38

- Loss of Immunological tolerance to self-liver antigen

TRIGGERS

- HAV, HBV, HCV, EBV
- Associated with:
 - IBD
 - Celiac sprue(anti-tTG)
 - MPGN
 - Hashimoto thyroiditis (anti-TPO antibodies)
 - Other antibodies- Anti-microsomal antibodies, anti-thyroglobulin antibodies

TYPE 1 AUTOIMMUNE HEPATITIS	TYPE 2 AUTOIMMUNE HEPATITIS
• Young female • North American descent • Anti-nuclear antibody present • Anti-smooth muscle antibody (A-SMA) present, directed against actin component of smooth muscle in blood vessels of liver • Anti-soluble liver antigen • X-ANCA: Atypical ANCA	• Young female/children • European • No anti-nuclear antibody present • Anti-LKM₁, antibody directed against cytochrome P₄₅₀ system • Anti-liver cytosol antibody (against cytoplasm of hepatocytes)

ANTIBODIES ASSOCIATED WITH INFECTIONS

Anti LKM 1	Associated with HCV
Anti LKM 2	Drug-induced hepatitis
Anti LKM 3	Associated with HDV

- Anti-HCV antibody against Hepatitis C is not sufficient for diagnosis
 - **PCR** needs to be done

CLINICAL FEATURES

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- Asymptomatic
- Symptomatic



- Fatigue, Anorexia, Acne, Amenorrhea
- Course (Waxing and waning course)
- Sjögren's syndrome (Dry eyes and mouth)
- Erythema nodosum: Purplish nodule present on the shin of the patient
- Cirrhosis: Can lead to ascites, portal hypertension, ↑ risk of HCC
→ 40% chance of mortality in 6 months if condition is undiagnosed

WORK UP

- LFT
 - Serum bilirubin ↑ (values are not static)
 - SGPT ↑
 - Serum ALP: N/↑
 - Serum albumin: ↓
 - Serum globulin: ↑↑ due to ↓ in catabolism
- Viral serology: **Negative**
- PT: normal in the initial part of illness and then ↑
- ANA, Anti-SMA, Anti-LKM₁, Antibody
- **Liver Biopsy**: IOC
 - Interface hepatitis
 - **Rosette formation** due to regenerating hepatocytes (Pseudo rosette formation in retinoblastoma, neuroblastoma)

DIAGNOSTIC CRITERIA FOR AUTOIMMUNE HEPATITIS

00:11:47

- Excluded causes
 - Viral insult
 - Drugs
 - Alcohol
 - Genetic diseases (α -1 AT deficiency, Wilson disease, Menke's disease)
- Include features
 - Antibodies panel
 - Biopsy

MANAGEMENT

- **Steroids + Azathioprine**

MCQ

- Q. Which of the following autoantibodies is most commonly associated with Type 1 Autoimmune Hepatitis?
- Anti-LKM1
 - Anti-liver cytosol-1 antibody
 - Anti-smooth muscle antibody
 - Anti-LKM3

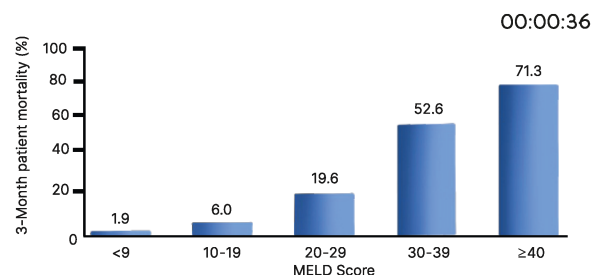
Ans: (c)



CIRRHOSIS AND ITS COMPLICATIONS

INTRODUCTION

- Sources used to **assess mortality risk** in cirrhosis:
 - MELD score** (Model for End-Stage Liver Disease)
 - C: Creatinine
 - B: Bilirubin
 - I: INR/PT
 - MELD score range: 6-40 (↑ score → ↑ mortality risk)
 - To enlist for Orthotopic Liver Transplantation: **≥17** (United Network for Organ Sharing- UNOS)
 - MELD-Na**: parameters are:
 - Bilirubin
 - INR
 - Creatinine
 - Serum sodium



MELD SCORE	MORTALITY
20-29	20%
30-39	50%

- 2.5x ↑ in mortality with ↑ in MELD score

(If dialysis was performed in the past 7 days → **creatinine is reset** to 4 mg/dL)

- PELD score** parameters:
 - Bilirubin
 - INR
 - Albumin
 - Age
 - Growth failure
- Child-Pugh score** parameters (each graded 1-3 points)
 - Bilirubin
 - INR
 - Albumin
 - Ascites
 - Asterixis: due to ammonia intoxication

(**value ≥7**: Enlist the patient for **orthotopic liver transplantation**)

CLASS	
A	5-6: Compensated cirrhosis
B	7-9: Decompensated cirrhosis
C	>10: ESLD

	1	2	3
ENCEPHALOPATHY	None	Grade 1-2 (or precipitant induced)	Grade 3-4 (or chronic)
ASCITES	None	Mild to Moderate (diuretic responsive)	Severe (diuretic refractory)
BILIRUBIN (MG/DL)	<2	2-3	>3
ALBUMIN (G/DL)	>3.5	2.8-3.5	<2.8
INR	<1.7	1.7-2.3	>2.3

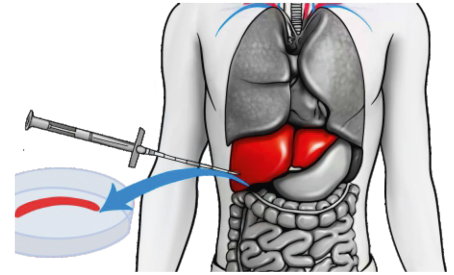
Child-Turcotte-Pugh Class obtained by adding a score for each parameter (total points)

Class A = 5 to 6 points (least severe liver disease)

INVASIVE TESTING FOR CIRRHOSIS

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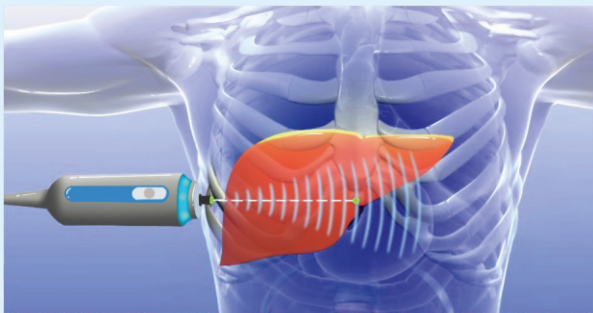
- Liver biopsy:
 - Most accurate test to assess the **severity of cirrhosis**
 - Check coagulogram before biopsy; if abnormal, administer Fresh Frozen Plasma (FFP) to normalize
 - Biopsy site: **8th intercostal space, mid-axillary line**
 - Post-procedure: Patient lies on right side (**right lateral decubitus**) to apply pressure and ↓ local bleeding
 - Scoring used
 - Metavir
 - Ishak



METAVIR SCORE SYSTEM	FIBROSIS STAGE
F0	No fibrosis can be detected
F1	Fibrosis exists with the expansion of portal zones
F2	Fibrosis exists with the expansion of most portal zones and occasional bridging
F3	Fibrosis exists with the expansion of most portal zones, marked bridging, and occasional nodules
F4	Presence of Cirrhosis

NON-INVASIVE TESTING FOR CIRRHOSIS

TRANSIENT ELASTOGRAPHY (FIBROSCAN)



Fibroscan

- >7.3 kPa: Advanced fibrosis
- >15kPa: Cirrhosis
- Reproducibility and accuracy of this is as good as liver biopsy

FIB-4 SCORE

- Factors taken into consideration
 - Age
 - AST
 - ALT
 - Platelet count
- In case of advanced fibrosis, advanced liver diseases
 - spleen size ↑ (hypersplenism), ↓ **platelet count**